Gimbal Control Node: Mathematical Model and Control Law

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1 System Overview

The 2–axis gimbal consists of two orthogonal brushless motors that control the roll (ϕ) and pitch (θ) of a platform. An IMU located at the intersection of the gimbal axes measures real–time roll, pitch, and yaw angles. The motors are controlled via CAN bus using the `0xA4` "Absolute Position Closed–Loop" command.

- IMU CAN ID: $0x100$
- Roll motor ID: m_ϕ (e.g. $0x142$)
- Pitch motor ID: m_θ (e.g. $0x143$)

2 IMU Data Decoding

Each IMU frame encodes roll, pitch, and yaw in centi–degrees (cd). The gimbal node converts to radians as:

$$\phi(t) = \frac{\pi}{18000}\, r_{cd}(t), \tag{1}$$
$$\theta(t) = \frac{\pi}{18000}\, p_{cd}(t), \tag{2}$$
$$\psi(t) = \frac{\pi}{18000}\, y_{cd}(t). \tag{3}$$

These values are published to the ROS 2 topic `/gimbal/imu_rpy_rad`.

3 Control Objectives

The controller maintains the platform near the desired reference angles

$$\phi_{\text{ref}}, \quad \theta_{\text{ref}}$$

(typically zero for self–balancing).

The gimbal angles are measured by the IMU and the control law produces commanded motor angles q_1, q_2 (for roll and pitch motors respectively).

4 Error Dynamics and PD Control

At each control cycle (t_k), the node computes the roll and pitch errors:

$$e_\phi(t_k) = \phi_{\text{ref}} - \phi(t_k), \tag{4}$$
$$e_\theta(t_k) = \theta_{\text{ref}} - \theta(t_k). \tag{5}$$

Angular rates are approximated using backward differences:

$$\dot{\phi}(t_k) = \frac{\phi(t_k) - \phi(t_{k-1})}{\Delta t}, \tag{6}$$
$$\dot{\theta}(t_k) = \frac{\theta(t_k) - \theta(t_{k-1})}{\Delta t}. \tag{7}$$

The PD control torques (or equivalent angular accelerations) are:

$$u_\phi(t_k) = K_{p,\phi}\, e_\phi(t_k) - K_{d,\phi}\, \dot{\phi}(t_k), \tag{8}$$
$$u_\theta(t_k) = K_{p,\theta}\, e_\theta(t_k) - K_{d,\theta}\, \dot{\theta}(t_k). \tag{9}$$

5 Integration to Platform Commands

The control signals are integrated to obtain the desired platform angles (the platform command in task space):

$$\phi_{\text{cmd}}(t_k) = \phi_{\text{cmd}}(t_{k-1}) + u_\phi(t_k)\, \Delta t, \tag{10}$$
$$\theta_{\text{cmd}}(t_k) = \theta_{\text{cmd}}(t_{k-1}) + u_\theta(t_k)\, \Delta t. \tag{11}$$

The commands are limited to a maximum range and slew rate:

$$|\phi_{\text{cmd}}| \le \phi_{\max}, \qquad |\theta_{\text{cmd}}| \le \theta_{\max}, \tag{12}$$
$$|\dot{\phi}_{\text{cmd}}| \le \dot{\phi}_{\max}, \qquad |\dot{\theta}_{\text{cmd}}| \le \dot{\theta}_{\max}. \tag{13}$$

6 Mapping to Motor Coordinates

Each motor has a zero offset ($q_{1,0}, q_{2,0}$) and a direction sign ($\sigma_\phi, \sigma_\theta$) depending on wiring orientation.

The commanded motor angles in radians are:

$$q_1^* = q_{1,0} + \sigma_\phi\, \phi_{\text{cmd}}, \tag{14}$$
$$q_2^* = q_{2,0} + \sigma_\theta\, \theta_{\text{cmd}}. \tag{15}$$

These are published as a ROS 2 topic:

$$\texttt{/gimbal/motor\_targets\_rad} = \begin{bmatrix} q_1^* \\ q_2^* \end{bmatrix}.$$

7 CAN Command Encoding

Motor commands are transmitted via CAN with the "Absolute Position Closed–Loop" frame (`0xA4`):

Byte	Symbol	Description
0	0xA4	Command ID
1	0x00	Reserved
2–3	$v_{\max}$	Max speed (1 dps/LSB)
4–7	p_{cmd}	Position (0.01°/LSB, little–endian)

Conversion from radians to centi–degrees:

$$p_{\mathrm{cmd}} = \mathrm{round}\left(q^* \frac{18000}{\pi}\right). \tag{16}$$

Thus the transmitted frame for the roll motor is:

$$\mathrm{CAN\ ID} = 0x140 + \mathrm{motor\_roll\_id}, \tag{17}$$

$$\mathrm{DATA} = [0xA4,\ 0x00,\ v_{\max}^{(L)},\ v_{\max}^{(H)},\ p_0,\ p_1,\ p_2,\ p_3]. \tag{18}$$

8 Overall Closed Loop

The complete flow is illustrated as:

$$\boxed{\begin{array}{ccc} (\phi_{\mathrm{ref}}, \theta_{\mathrm{ref}}) & \xrightarrow{\text{compare}} & (e_\phi, e_\theta) \\ & \searrow \text{PD} & \downarrow \text{integrate \& slew limit} \\ & & (\phi_{\mathrm{cmd}}, \theta_{\mathrm{cmd}}) \\ & & \downarrow \text{map to motors} \\ & & (q_1^*, q_2^*) \xrightarrow{\text{CAN 0xA4}} \text{Motors} \end{array}}$$

IMU feedback continuously updates (ϕ, θ), closing the loop to maintain $\phi \to \phi_{\mathrm{ref}}$ and $\theta \to \theta_{\mathrm{ref}}$.

9 Equilibrium Condition

At steady state,

$$e_\phi = \phi_{\mathrm{ref}} - \phi = 0, \quad e_\theta = \theta_{\mathrm{ref}} - \theta = 0, \tag{19}$$

which implies the platform is level and the motor target angles equal the zero offsets:

$$q_1^* = q_{1,0}, \quad q_2^* = q_{2,0}.$$

10 Safety Constraints

- IMU timeout: skip control if no update for $t_{\mathrm{timeout}} = 0.5$ s.
- Slew limiting prevents aggressive steps: $|\Delta q^*| \le \dot{q}_{\max} \Delta t$
- Control disabled if `enable_can_command = false`.

11 Summary of Key Parameters

Parameter	Symbol	Meaning
$K_{p,\phi}, K_{p,\theta}$	–	Proportional gains
$K_{d,\phi}, K_{d,\theta}$	–	Derivative gains
$\phi_{\max}, \theta_{\max}$	deg	Angle limits
$\dot{\phi}_{\max}, \dot{\theta}_{\max}$	deg/s	Slew rate limits
$\sigma_\phi, \sigma_\theta$	± 1	Sign convention
$q_{1,0}, q_{2,0}$	rad	Motor zeros
$v_{\max}$	dps	Motor command speed limit